

Deep Learning-Based Animal Re-Identification for Non-Invasive Wildlife Monitoring and Conservation

Matej Marić

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Motivation: Why a Species-Agnostic Pipeline?

- Unified pipeline across species
- Minimal supervision, practical runtime
- Re-ID as a building block for population estimation

Datasets: WildlifeReID-10k (7 Used Here)

WildlifeReID-10k: 37 datasets, 33 species, 140k images, 10.7k individuals.
Evaluated on 7 represe (closed-set).

Dataset	# images	# individuals	split type
ATRW*	5,415	182	Original MD split
CZoo*	4,662	71	Original MD split
SeaIID*	2,080	57	Original MD split
CowDataset	1,485	13	Time-aware split
Chicks4FreeID	1,146	50	Similarity-aware split
SeaStarReID2023	2,187	95	Time-aware split
ELPephants	2,078	276	Random split

Datasets: WildlifeReID-10k (7 Used Here)



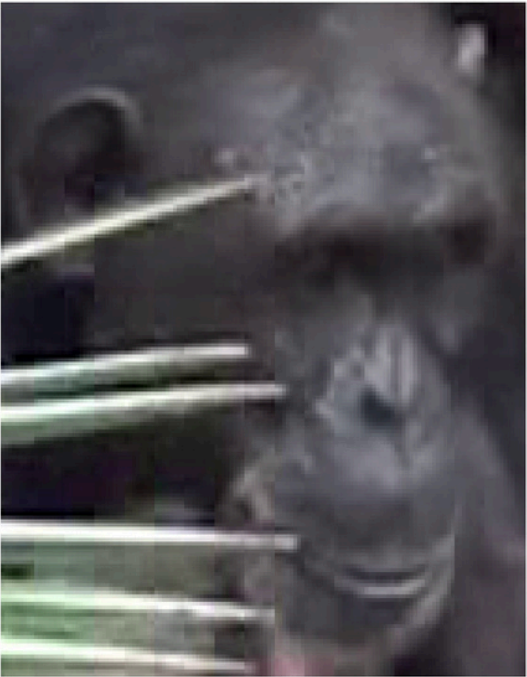
atrw



cowdataset



elpephants



CZOO



chicks4freeid



sealid



seastarreid2023

Datasets: Data Leaks

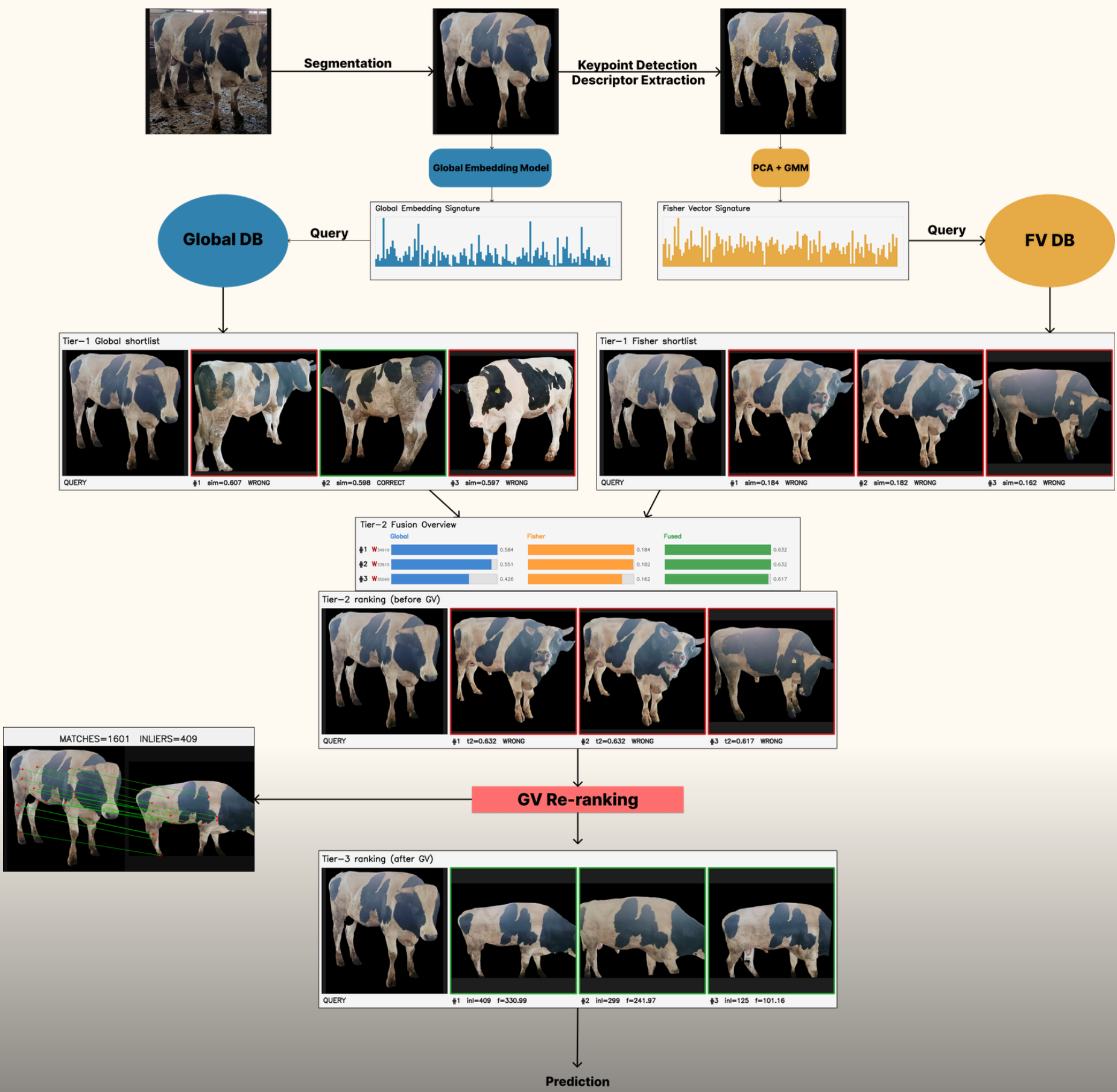
- Time-aware splits
- Similarity-aware splits
- MD-train split alignment (ATRW/CZoo/SealID*)



Classification Funnel: Retrieve -> Rerank -> Verify

- Tier-1: retrieve candidates (Global + Fisher) and take the union
- Tier-2: calibrate and fuse scores into same-ID probability
- Tier-3: geometric verification (keypoints + RANSAC) on the shortlist

Pipeline Overview (Full)

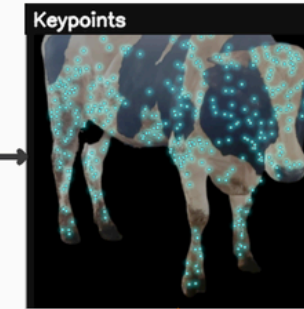
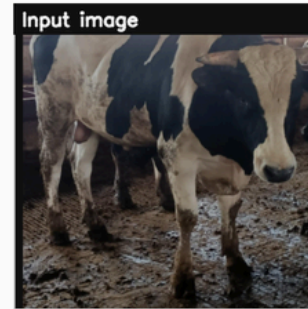


Tier 1: Candidate Retrieval (Concept)

- Two fast signals: Global embeddings + Fisher vectors
- Take top-K from each and form an ordered union (deduped)
- Goal: high-recall shortlist for expensive stages
- Background removal helps stabilize local evidence (optional)

Tier 1: Candidate Retrieval (Visualization)

Tier-1 Candidate Retrieval: Global + Fisher $\rightarrow$ Union



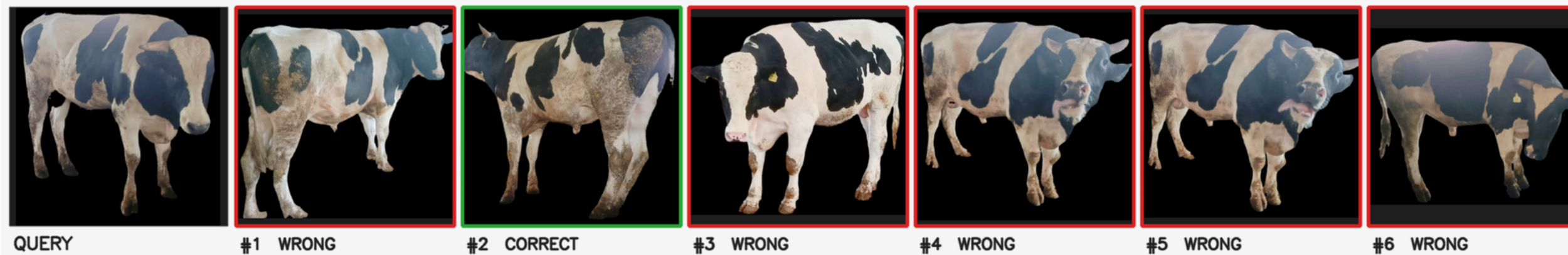
Tier-1 Global shortlist



Tier-1 Fisher shortlist



Tier-1 Union shortlist



Tier 2: Calibration and Fusion (Concept)

- Raw similarities are not comparable across modalities
- Per-dataset calibration: score $\rightarrow P(\text{same} \mid \text{score})$
- Fuse calibrated Global + Fisher (simple average)
- Output: Tier-2 reranking of the union shortlist

Tier 3: Geometric Verification (Concept)

- Run only on a small shortlist (most expensive stage)
- Match local keypoints and estimate geometric consistency
- Inlier count provides strong evidence against false matches
- Especially helpful on hard datasets (e.g., ELPephants)

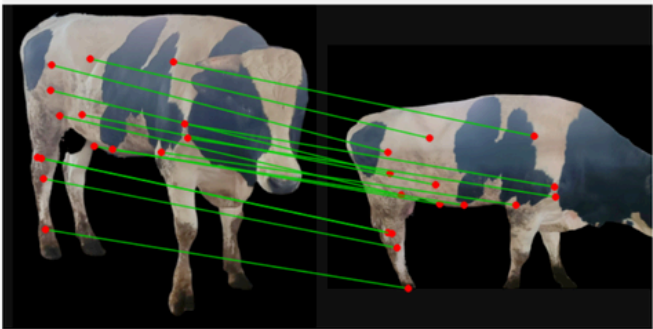
Tier 3: Geometric Verification (Visualization)

Tier-3: Geometric Verification (GV) re-ranks Tier-2

Tier-2 ranking (before GV)



MATCHES=1601 INLIERS=409



Tier-3 ranking (after GV)



Classification Results (Mean over 7 datasets)

Metric	Global	WildFusion	This thesis
Top-1 (%)	71.75	88.18	86.73
Top-5 (%)	79.58	91.91	90.70
F1 (%)	71.51	87.35	85.70
Runtime (min)	0.06	214.52	42.53

- WildFusion: strong calibrated fusion baseline
- GV is expensive -> applied only on a shortlist
- Key point: near-WildFusion accuracy at ~5x lower runtime

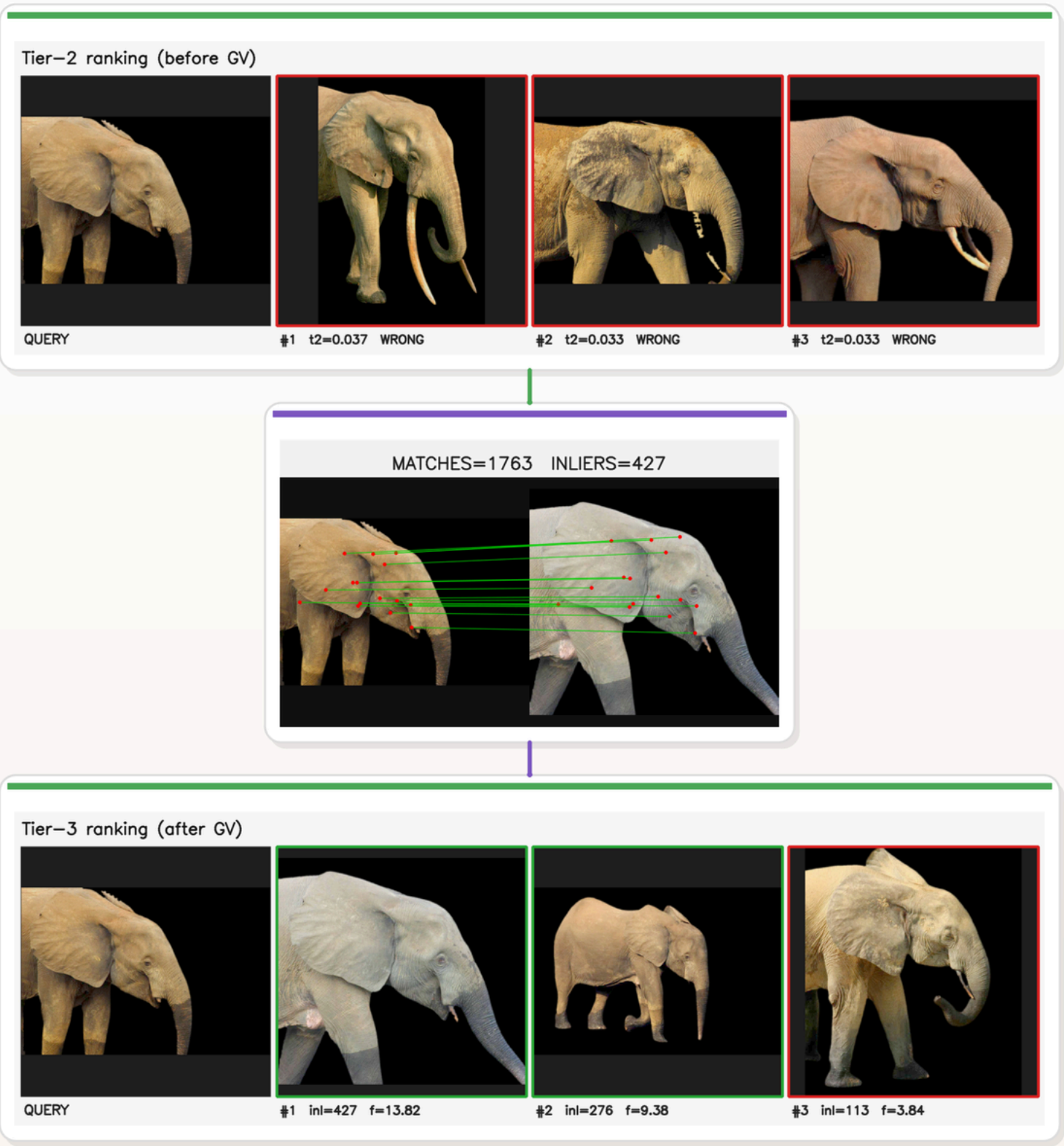
Classification Results (Selected Datasets)

Dataset	Metric	Global	WildFusion	Fisher	G+F	F+GV	G+F+GV
ELPephants	Top-1	13.66	49.31	20.79	21.58	46.73	54.46
	Top-5	21.78	57.03	31.49	32.48	68.12	64.95
	F1	11.58	45.19	18.21	19.26	42.87	49.71
SeaIID*	Top-1	78.42	97.60	65.23	79.38	81.53	85.13
	Top-5	79.62	98.56	80.58	82.73	89.69	88.97
	F1	77.86	97.67	64.59	78.47	80.58	84.40
SeaStarRelD2023	Top-1	47.91	80.47	68.84	62.79	77.21	77.21
	Top-5	73.49	90.70	85.58	82.79	85.58	85.58
	F1	45.71	79.09	67.92	60.70	76.40	76.85

GV gain (ELPephants Top-1): 21.58 (G+F) -> 54.46 (G+F+GV).

Classification Results

Tier-3: Geometric Verification (GV) re-ranks Tier-2



Population Counting: HTL-NIS (Idea)

- Goal: estimate #individuals without full clustering
- Model images as clique graph $\rightarrow$ connected components
- Query only a small number of pairs (oracle: same/different)
- Use Nested Importance Sampling to reduce variance

$$K = \sum_u 1 / (1 + d(u))$$

N vertices, M neighbors

Counting Drawbacks: Human Error Matters

- Pair labels are highly imbalanced (mostly 'different')
- False positives merge identities -> underestimation collapse
- Confidence intervals capture sampling variance, not bias
- Mitigation: positive confirmation (accept 'same' only after K votes)

Population Results (K=2): Full Table

	ATRW	CowDataset	Chicks4FreeID	CZoo	ELPephants	SealID	SeaStarReID2023
#images	5415	1388	1086	2109	2078	2080	2077
GT	182	12	48	24	274	57	91
p=0.00	278 [132, 425]	12 [11, 13]	50 [43, 58]	24 [23, 26]	334 [267, 401]	56 [45, 68]	106 [85, 128]
0.02	212 [124, 300]	13 [11, 14]	50 [42, 57]	25 [23, 27]	320 [259, 381]	65 [47, 83]	106 [87, 126]
0.05	178 [116, 240]	13 [12, 14]	47 [41, 54]	26 [24, 27]	178 [157, 199]	53 [43, 62]	93 [79, 106]
0.10	72 [58, 86]	13 [12, 14]	39 [34, 43]	25 [23, 27]	78 [73, 83]	45 [37, 54]	59 [52, 66]
0.15	39 [34, 45]	12 [11, 14]	27 [24, 30]	22 [20, 24]	40 [38, 43]	33 [29, 38]	37 [33, 41]
0.30	11 [10, 12]	8 [7, 9]	10 [9, 11]	13 [11, 14]	11 [10, 12]	13 [11, 15]	11 [10, 12]

Bold: GT inside the 95% confidence interval.

Population Results (K=1): Full Table

	ATRW	CowDataset	Chicks4FreeID	CZoo	ELPephants	SealID	SeaStarReID2023
#images	5415	1388	1086	2109	2078	2080	2077
GT	182	12	48	24	274	57	91
p=0.00	278 [132, 425]	12 [11, 13]	50 [43, 58]	24 [23, 26]	342 [298, 386]	56 [45, 68]	106 [85, 128]
0.02	40 [33, 46]	10 [9, 11]	25 [22, 28]	19 [17, 20]	42 [41, 44]	29 [25, 34]	36 [32, 40]
0.05	18 [16, 20]	8 [7, 9]	15 [13, 16]	13 [12, 15]	19 [18, 20]	17 [15, 20]	18 [16, 20]
0.10	10 [9, 11]	6 [5, 7]	9 [8, 10]	9 [8, 10]	10 [9, 10]	11 [9, 12]	10 [9, 11]
0.15	7 [6, 7]	5 [4, 5]	6 [5, 7]	7 [6, 8]	7 [6, 7]	8 [7, 8]	7 [6, 7]
0.30	3 [3, 4]	3 [2, 4]	3 [3, 4]	4 [3, 5]	3 [3, 4]	4 [3, 5]	3 [3, 4]

Bold: GT inside the 95% confidence interval.

Conclusions

- Species-agnostic 3-tier Re-ID works across multiple benchmarks
- Global + Fisher + GV gives strong accuracy/runtime trade-off
- HTL-NIS is promising, but very sensitive to human error

Future Work

- Validate counting with real annotators (not simulated error)
- Better global backbone and stronger local matching for hard datasets
- Link photographed-individual counts to true abundance/density

Thank you Questions?

(a) Original Input



(b) GroundingDINO Detection



(c) SAM 2 Segmentation



(d) Final Soft-Masked Output

